A16729W1

SECOND PUBLIC EXAMINATION

Honour School of Physics Part B: 3 and 4 Year Courses

Honour School of Physics and Philosophy Part B

B5: GENERAL RELATIVITY

MICHAELMAS TERM 2020

Friday, 09 October

Opening time: 09.30 am (BST)

You have two hours and thirty minutes to complete the paper and upload your answer file

Answer **two** *questions.*

Start the answer to each question on a fresh page.

A list of physical constants and conversion factors is available at the Physics website.

You are permitted to use calculators.

The numbers in the margin indicate the weight that the Examiners expect to assign to each part of the question.

Page 2 contains physics formulae and data for this paper.

The questions start on page 3.

Expressed in terms of the Christoffel symbols $\Gamma^{\alpha}_{\beta\gamma}$, the Riemann tensor has components

$$R^{\gamma}{}_{\alpha\beta\delta} = \frac{\partial}{\partial x^{\delta}}\Gamma^{\gamma}_{\alpha\beta} - \frac{\partial}{\partial x^{\beta}}\Gamma^{\gamma}_{\alpha\delta} + \Gamma^{\epsilon}_{\alpha\beta}\Gamma^{\gamma}_{\delta\epsilon} - \Gamma^{\epsilon}_{\alpha\delta}\Gamma^{\gamma}_{\beta\epsilon}.$$

The Ricci tensor is defined as $R_{\alpha\beta} \equiv R^{\gamma}{}_{\alpha\beta\gamma}$. When expressed in a basis in which the metric tensor $g_{\alpha\beta}$ is diagonal, it can be written as

$$R_{\alpha\beta} = \frac{1}{2}\frac{\partial^2}{\partial x^{\alpha}\partial x^{\beta}}\ln|g| - \frac{\partial\Gamma^{\gamma}_{\alpha\beta}}{\partial x^{\gamma}} + \Gamma^{\delta}_{\alpha\gamma}\Gamma^{\gamma}_{\beta\delta} - \frac{1}{2}\Gamma^{\gamma}_{\alpha\beta}\frac{\partial}{\partial x^{\gamma}}\ln|g|,$$

where $|g|$ is the modulus of the determinant of $g_{\alpha\beta}$.

The Einstein equation is

$$R_{\alpha\beta} - \frac{1}{2}g_{\alpha\beta}R + \Lambda g_{\alpha\beta} = -\frac{8\pi G}{c^4}T_{\alpha\beta},$$

where $R \equiv g^{\alpha\beta}R_{\alpha\beta}$ is the Ricci scalar, Λ is the cosmological constant and $T_{\alpha\beta}$ is the energy–momentum tensor.

For the line element

$$\mathrm{d}s^2 = -c^2\,\mathrm{d}t^2 + R^2(t)\left[\frac{\mathrm{d}r^2}{1-kr^2} + r^2\,\mathrm{d}\theta^2 + r^2\sin^2\theta\,\mathrm{d}\phi^2\right]$$

the Einstein equation can be written as the pair of equations

$$\begin{aligned}\ddot{R} &= -\frac{4\pi G}{3}\left(\rho + \frac{3p}{c^2}\right)R + \frac{1}{3}\Lambda c^2 R,\\ \dot{R}^2 &= \frac{8\pi G}{3}\rho R^2 + \frac{1}{3}\Lambda c^2 R^2 - c^2 k,\end{aligned}$$

where ρ and p are the rest-frame density and pressure, respectively.

1. (a) In some curved spacetime with coordinates x^μ, use the locally free-falling coordinates ξ^μ to define the affine connection $\Gamma^\lambda_{\mu\nu}$. For a vector V^μ, define the covariant derivative with respect to the coordinates x^μ in terms of the ordinary partial derivative and the affine connection. [4]

(b) Is the connection coefficient $\Gamma^\lambda_{\mu\nu}$ a tensor? Write down a proof for your answer. [6]

(c) The covariant derivative of the tensor T^μ_σ is given by

$$T^\mu_{\sigma;\tau} = \frac{\partial T^\mu_\sigma}{\partial x^\tau} + \Gamma^\mu_{\nu\tau} T^\nu_\sigma - \Gamma^\nu_{\sigma\tau} T^\mu_\nu .$$

Investigate the commutation rule for covariant derivatives by evaluating $V^\mu_{;\sigma;\tau}$, where $_{;\sigma}$ symbolizes the covariant derivative with respect to x^σ. Now subtract $V^\mu_{;\tau;\sigma}$ (the indices τ, σ have been swapped) from your result, and show that $V^\mu_{;\sigma;\tau} - V^\mu_{;\tau;\sigma} = R^\mu_{\lambda\sigma\tau}\, V^\lambda$. [6]

(d) Argue why the Riemann curvature tensor $R^\mu_{\lambda\sigma\tau}$ must therefore be a tensor. [5]

(e) Write down the relationships between $R^\mu_{\lambda\sigma\tau}$, $R_{\lambda\mu\sigma\tau}$ (the fully covariant form of the tensor), the Ricci tensor $R_{\mu\nu}$ and the curvature scalar R, without going into details. What symmetries are known for $R_{\lambda\mu\sigma\tau}$? [4]

2. (a) Convert the Minkowski line element to that of a uniformly accelerated reference frame with coordinates (t', x', y', z'), given the following relationships:

$$ct = \left(\frac{c^2}{g} + z'\right) \sinh\left(\frac{gt'}{c}\right)$$

$$z = \left(\frac{c^2}{g} + z'\right) \cosh\left(\frac{gt'}{c}\right) - \frac{c^2}{g}$$

$$y = y'$$

$$x = x',$$

where g is a constant acceleration. [4]

(b) Write down the condition under which this corresponds to a transformation to a uniformly accelerated field in Newtonian mechanics, and show explicitly why that is the case. [4]

(c) In the primed frame, calculate the proper time at coordinates $z' = 0$ and $z' = h$, and show that a clock at rest $z' = h$ runs fast compared to a clock at rest at $z = 0$ by a factor

$$1 + \frac{gh}{c^2}.$$

[3]

(d) What is gravitational time dilation and how is it analogous to the time dilation described here? [6]

(e) Compute the invariant acceleration $a = (a_\mu a^\mu)^{1/2}$, where $a^\mu = \frac{\mathrm{d}^2 x^\mu}{\mathrm{d}\tau^2}$, and show that it is not the same at $z' = h$ and $z' = 0$. [8]

3. (a) A spaceship is positioned in a circular orbit around a black hole. The metric around the black hole is given by

$$-c^2\,\mathrm{d}\tau^2 = -\left(1-\frac{2GM}{c^2 r}\right)c^2\,\mathrm{d}t^2 + \left(1-\frac{2GM}{c^2 r}\right)^{-1}\mathrm{d}r^2 + r^2\,\mathrm{d}\theta^2 + r^2\sin^2\theta\,\mathrm{d}\phi^2.$$

The radius of the orbit is $R_\mathrm{o} = 11GM_\mathrm{bh}/c^2$, where M_bh is the mass of the black hole. What is the angular velocity $\frac{\mathrm{d}\phi}{\mathrm{d}\tau}$ as measured at the spaceship? Use the appropriate geodesic equation to compute $\frac{\mathrm{d}\phi}{\mathrm{d}t}$ and the metric to compute $\frac{\mathrm{d}t}{\mathrm{d}\tau}$. [6]

(b) What is the period of the orbit for an observer at infinity? [3]

(c) What is the period of the orbit measured by a clock on the spaceship? [3]

Now consider two distant black holes of mass $M_\mathrm{bh} = 10^6 M_\odot$ each. At some moment in time, they are orbiting each other in a circular orbit with a period of 5×10^4 s as measured from Earth.

(d) Show that their distance in units of Schwarzschild radii, $\frac{2GM_\mathrm{bh}}{c^2}$, is given by:

$$R_s = \left(\frac{T^2 c^6}{32\pi^2 G^2 M_\mathrm{bh}^2}\right)^{\frac{1}{3}},$$

where T is the orbital period. [6]

The gravitational radiation quadrupole formula at a large distance r is given by

$$\bar{h}^{ij} = \frac{2G}{c^6 r}\frac{\mathrm{d}^2 I^{ij}}{\mathrm{d}t'^2},$$

where the quadrupole moment tensor I^{ij} is given by $I^{ij} = M_\mathrm{bh}c^2 x_1^i x_1^j + M_\mathrm{bh}c^2 x_2^i x_2^j$.

(e) Assuming all the black hole mass to be concentrated at the centre of each object, and defining x_1^i and x_2^i as the Cartesian coordinates on the plane of the circular orbit of the two black holes, find the frequency of the gravitational wave emitted in Hz. [7]

4. (a) The source term in the gravitational field equations of General Relativity must be a tensor that describes the distribution of matter at each position in space-time. Consider an ideal relativistic fluid made of a distribution of non-interacting dust particles (no pressure, no forces between particles, no heat conduction and no viscosity in its instantaneous rest frame), each of rest mass m_0. This fluid can be characterized at each space-time position A by its bulk 3-velocity $\mathbf{u}$ (as measured in an inertial frame) and matter density at rest ρ, a scalar. By considering the fluid to be instantaneously at rest at A, and Lorentz transforming the density at rest ρ into a moving frame S' with Lorentz factor γ, explain why the source term in the field equations should be a rank-2 tensor. [5]

(b) Construct such a rank-2 tensor $T^{\mu\nu}$, using the scalar ρ and the bulk 4-velocity of the fluid, U^{μ} in the S' frame, and write down the $T^{\mu\mu}$ terms in terms of γ, ρ, c, and the 4-velocity of the fluid. [3]

(c) Now, instead of pressureless dust, consider a perfect fluid with pressure. In the instantaneous rest frame, $T^{\mu\nu}$ becomes

$$T^{\mu\nu} = \begin{pmatrix} \rho c^2 & 0 & 0 & 0 \\ 0 & P & 0 & 0 \\ 0 & 0 & P & 0 \\ 0 & 0 & 0 & P \end{pmatrix},$$

where P is the pressure of the fluid. Explain why the $i = j$ terms are indeed the pressure of the fluid. Why are the off diagonal terms all 0? [2]

(d) Now show that $T^{\mu\nu}$ in any inertial frame (Minkowski metric $\eta^{\mu\nu}$) is:

$$T^{\mu\nu} = (\rho + P/c^2)U^{\mu}U^{\nu} + P\eta^{\mu\nu}.$$

Why is this the unique form of $T^{\mu\nu}$? [6]

(e) Why is it possible to exchange $\eta^{\mu\nu}$ with $g^{\mu\nu}$ in the presence of gravity? [4]

(f) For the relativistic fluid described above, show that the acceleration $A^{\mu} = U^{\nu}U^{\mu}_{;\nu}$, where $_{;\mu}$ denotes the covariant derivative, is orthogonal to the velocity U^{μ}. [5]